Socialized Unlearning

Radu Loghin Paul Alexandru
Sapienza Università di Roma
raduloghin.1942544@studenti.uniroma1.it

Maggiore Andrea
Sapienza Università di Roma
maggiore.1947898@studenti.uniroma1.it

Abstract

The increasing demand for privacy, adaptability, and robustness in machine learning models has made selective unlearning an essential capability. By leveraging collaborative multi-agent interactions, this work introduces the concept of *Socialized Unlearning*, combining mechanisms such as **collaborative collaboratio**n and **reciprocal altruism** to balance stability and plasticity during the forgetting process. Experimental results demonstrate the framework's effectiveness in reducing the influence of target knowledge while preserving performance on non-target data, making us understand that with further future developments it could become an excellent unlearning technique.

1 Introduction

The rapid adoption of machine learning (ML) systems in real-world applications has raised critical challenges related to privacy, adaptability, and robustness. One such challenge is *selective unlearning*, the process of removing specific knowledge from a model without compromising its overall performance. This is particularly relevant in scenarios where data must be forgotten due to privacy regulations, outdated information, or ethical considerations.

Traditional unlearning approaches often rely on retraining the model from scratch, which is computationally expensive and impractical for large-scale systems. Furthermore, achieving a balance between forgetting specific knowledge and retaining useful information remains a significant challenge.

2 Related Work

The first phase of our project involved the reimplementation of the framework proposed in [1]. This foundational work introduces the concept of socialized learning, where collaborative interactions between multiple agents are leveraged to enhance knowledge sharing and correction in machine learning models. Our reimplementation allowed us to gain a deeper understanding of the techniques and principles underlying this approach, providing the basis for developing our unlearning framework.

The same paper [1] also helps us to find the intuition behind the implementation of this unlearning technique. One of the fundamental challenges in machine learning lies in rectifying incorrect knowledge acquired by models during training. In the context of unlearning, this issue becomes particularly critical when a model—analogous to a student—has internalized erroneous representations of specific classes. The role of the teacher, in this case, is to guide the model in forgetting the target class while simultaneously ensuring that the model's knowledge is steered in the correct direction. This collaborative dynamic between teacher and student aligns with the principles of socialized learning [1], where multi-agent interactions facilitate the manipulation and correction of targeted knowledge. In the following you can find how we set our environment:

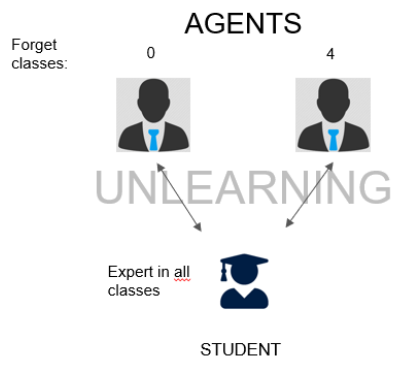


Figure 1: Unlearning environment

2.1 Proposed method explained

We propose an alternative version of the MASC framework[1], the Multi-Agent Socialized Unlearning Collaboration (MASUC).
This new framework is composed by the following modules:

Collaborative Unlearning

$$L^S = L^S_{\text{eras}}(p^S, T) + \delta_1 \sum_{i=1}^{N} L^S_{\text{KD}}(\hat{p}^S_i, p^t_i) + \delta_2 L^S_{\text{AL}}(p^S, \Delta)$$

Collaborative Collaboration

$$L^S = L^S_{\text{CE}}(p^S, p^{gt}) + \delta_1 \sum_{i=1}^{N} L^S_{\text{KD}}(\hat{p}^S_i, p^t_i) + \delta_2 L^S_{\text{AL}}(p^S, \Delta)$$

Reciprocal Unlearning

$$L^i_t = L^t_{\text{eras}}(p^i_t, T) + \delta_1 L^t_{\text{KD}}(\hat{p}^i_t, p^i_t) + \delta_2 L^t_{\text{AL}}(p^i_t, \Delta)$$

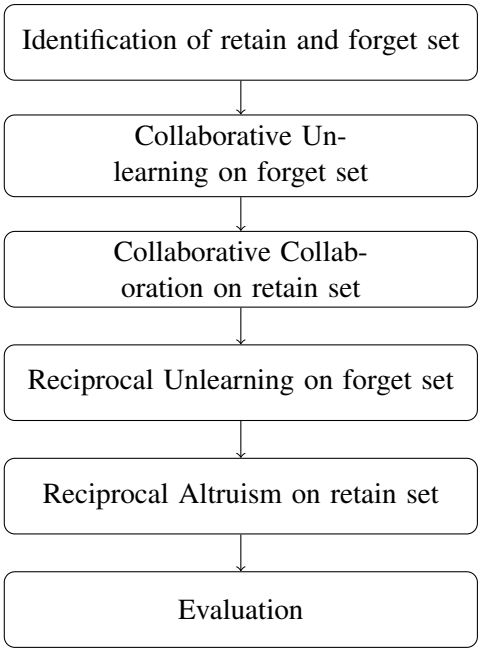
Reciprocal Altruism

$$L^i_t = L^t_{\text{CE}}(p^i_t, p^{gt}) + \delta_1 L^t_{\text{KD}}(\hat{p}^i_t, p^i_t) + \delta_2 L^t_{\text{AL}}(p^i_t, \Delta)$$

Now let us give a little explanation of the used formula:

- **Cross-Entropy Loss (L^S_{CE}, L^t_{CE}):** This term measures the discrepancy between the predictions of the model p^S / ${p_t}^i$ and the ground truth labels p^{gt}, ensuring that the model learns to classify correctly.
- **Knowledge Distillation Loss (L^S_{KD}, L^t_{KD}):** This component facilitates the transfer of knowledge from teacher models to the student and vice versa. $\hat{p}^S_i$ / $\hat{p}^i_t$ denotes the prediction of the classification head of the original teacher after receiving the characteristics of the grown student.
- **Alignment Loss (L^S_{AL}, L^t_{AL}):** This term ensures the alignment of the energy distributions p^S / p^i_t of the student with a target distribution Δ (dynamically computed based on the energy average of the output distributions), preserving the critical representations learned.
- **Erasure Loss (L^S_{eras}, L^t_{eras}):** This component increases the entropy of the target classes *T* to make the model predict wrong.

Now you can find in the following the proposed workflow of our unlearning method:



3 Dataset and benchmark

In this work, we evaluate our proposed socialized unlearning framework on standard dataset commonly used for image classification and knowledge distillation tasks. In the following, we provide an overview of the dataset and the benchmarking strategy used.

- **CIFAR-10:** The CIFAR-10 dataset is a widely-used benchmark for image classification, consisting of 60,000 32x32 color images categorized into 10 classes. The dataset is divided into 50,000 training samples and 10,000 test samples. Each class contains an equal number of images, ensuring balanced representation. For the unlearning part, we decided to compose the forget set by two classes: **0 (airplane)** and **4 (deer)**.

To assess the performance of our framework, we follow a comprehensive benchmarking strategy:

- **Metrics:** The performance is evaluated using standard metrics, including accuracy on the *retain set* and Retention Score.
- **Experimental Setup:** All experiments are conducted on a consistent hardware setup with Apple Silicon MPS GPU. We train our models for 50 epochs, using Adam as Optimizer and ReduceLROnPlateau as learning rate scheduler.

4 Experimental Results

For the first part of the project (reimplementation of the Socialized learning [1]) we got the following results: Figure 2 illustrates the loss trajectory of the student model. The loss demonstrates a sharp decline in the initial epochs, indicative of rapid knowledge acquisition, followed by a plateau as training progresses. This behavior aligns with expectations for effective learning models. Figure 3 shows the losses of the five teacher models over 50 epochs. While all teachers demonstrate a general trend of decreasing loss, fluctuations are evident, especially in the later epochs. These variations may result from differences in the teachers' assigned roles and the complexities of the data subsets they manage. Now let's see the results of the second part of our project: the unlearning. Figure 4 illustrates the loss behavior on the forget set for the student (*S*) and the two teachers (*T0* and *T1*) during the unlearning process. The forget set comprises the data points that the model is explicitly trained to unlearn, with the aim of reducing the student's reliance on these specific classes. As shown in the plot, the student (*S*) exhibits a noticeable increase in loss on the forget set across the epochs, indicating successful unlearning. This trend demonstrates that the student progressively forgets the target knowledge associated with the forget set. In contrast, the losses for the teachers (*T0* and *T1*) remain relatively stable throughout the process, which is consistent with their role as they already had unlearned their respective classes.

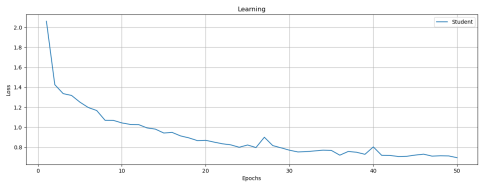


Figure 2: Student socialized learning

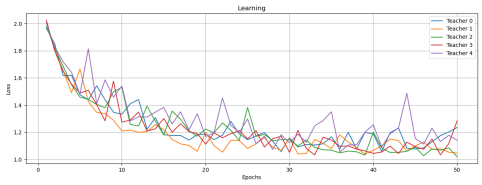


Figure 3: Teachers socialized learning

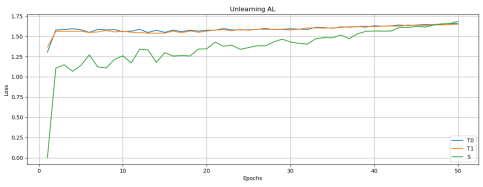


Figure 4: Unlearning loss over the epochs

Table 1 presents the pre-unlearning and post-unlearning accuracies, along with the computed Retention Score (RS) for the student and two teacher models. The Retention Score quantifies the ability of the models to preserve knowledge related to non-target classes after the unlearning process. The pre-unlearning accuracy for all models is approximately 72%, indicating comparable initial performance. Following the unlearning procedure, the post-unlearning accuracies decrease across all models, as expected, since unlearning aims to remove specific target class knowledge. However, the Retention Score highlights the models' ability to retain knowledge unrelated to the target class. The student model achieves a retention score of 0.900, indicating that 90% of its non-target knowledge remains intact after unlearning. Among the teacher models, Teacher 1 achieves the highest retention score (0.927), suggesting a slightly better preservation of non-target knowledge compared to Teacher 2 (0.875). This difference might reflect variations in the teachers' learning objectives or how they interact with the student during unlearning.

	Pre	Post	Retention Score
Student	72.32%	65.11%	0.900
Teacher 1	72.05%	66.80%	0.927
Teacher 2	72.26%	63.26%	0.875

Table 1: Accuracies pre and post unlearning

5 Conclusions and Future Work

In this work, we proposed a novel framework for socialized unlearning inspired by the principles of collaborative learning and reciprocal altruism. By leveraging the interplay between teachers and students, our method effectively guides models to forget undesired target knowledge while retaining crucial non-target information. However, there are several areas for improvement and exploration. First, the scalability of the framework to larger datasets and more complex architectures requires further investigation. Second, integrating advanced energy alignment mechanisms

and exploring alternative optimization strategies could enhance the unlearning process. In future work, we could refine the current framework by incorporating adversarial techniques to strengthen the robustness of unlearning. Moreover, a deeper exploration of ethical and privacy-related aspects of unlearning is critical, particularly in sensitive applications where data erasure and fairness are paramount. Finally, investigating the interplay between unlearning and continual learning could yield new insights into lifelong machine learning systems.

References

[1] Xinjie Yao, Yu Wang, Pengfei Zhu, Wanyu Lin, Jialu Li, Weihao Li, and Qinghua Hu. Socialized learning: Making each other better through multi-agent collaboration. In *Forty-first International Conference on Machine Learning*, 2024. 1, 2, 3